

Pramit Pal

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Education

Birla Institute of Technology and Science, Pilani
B.E. Electronics and Communication Engineering

2023 – Present
CGPA: 9.30/10

Research and Publications

Inverse Design of Realizable Metasurface-Based Absorbers Using Improved Conditioning and Diversity-Enhanced Progressively Growing GANs

Scientific Reports, Nature Portfolio. Accepted for publication. arXiv:2606.05849

- Co-authored a generative framework for inverse design of fabrication-aware metasurface absorbers, with continuous FiLM conditioning, electromagnetic-surrogate guidance, and determinantal-point-process diversity regularisation.
- Addressed the challenge of generating geometrically diverse, physically realizable designs that match continuous spectral targets; reported designs span 2–18 GHz.
- Reported evaluation results include 0.0052 average MSE, 0.8730 diversity score, 0.8533 band-alignment accuracy, and an 89.57% valid-EM-design rate.

Experience

D. E. Shaw

Jun – Jul 2026

Technology Development Intern, Quant Systems Development

- Evaluated identity, firewall access, DNS, proxy routing, and TLS requirements for extending service discovery into a network-restricted domain.
- Developed an eBPF program to integrate a virtual machine’s Layer 3 network space into the firm’s grid network for userspace testing of network technologies.

Selected Engineering Projects

openEMS-CUDA: Batched GPU Electromagnetic Simulation

2026

Independent systems and GPU engineering

C++, CUDA, Python

- Built a CUDA-accelerated batch FDTD engine for running openEMS parameter sweeps and inverse-design simulations, with fused kernels, CUDA Graph replay, and batched port-monitor processing.
- Designed runners and Python orchestration for ragged grid dimensions, independent early stopping, field snapshots, and in-memory simulation batches.
- Validated bit-for-bit E-field parity against the per-simulation GPU engine. A 16-configuration dipole benchmark on an RTX 4050 Laptop GPU reached 15.5 simulations per second; the reported I time-series relative error was at most 2.2×10^{-7} .
- Repository: github.com/ProAltro/openems-cuda.

Making Computation Fit: Edge AI and CXL-Backed MoE

2026

Hardware and systems research

SystemVerilog, FPGA, CXL, Python

- Developed a streaming power-forecasting pipeline in SystemVerilog combining engineered features, quantized temporal-convolution inference, and a tree ensemble for six forecast horizons.
- Optimised the TCN backbone for a Zynq-7020: Vivado 2019.2 out-of-context synthesis reduced LUTs from 87,430 to 51,506 (41%) and flip-flops from 199,188 to 65,029 (67%), using 28 BRAM36 blocks and 48 DSPs. This is synthesis data, not measured board throughput.
- Built reproducible CXL memory-placement and serving experiments for Qwen3-30B-A3B. In simulation, the 56-GiB aggregate model exceeded 48 GiB of aggregate HBM; CXL expansion enabled an 8/8-request run with about 27 GiB per GPU remotely placed.
- Compared against the matched ideal-HBM reference: the simulated CXL run had $27.05\times$ mean TTFT and $213.05\times$ mean TPOT. The study distinguishes making a model fit from improving serving performance.

Nebula GPU Interconnect Design Competition

Second Place, Hardware Design

Aug – Sep 2025

SystemVerilog, NoC

- Designed a network-on-chip for up to 64 GPUs with custom deadlock-free arbitration and AXI/CHI-compatible interconnect modules.
- Built SystemVerilog test benches for traffic patterns representative of shared GPU-training workloads; placed second in the hardware-design category of the Astera Labs Nebula competition.

Team Anant Satellite Simulation (sat-sim)

Satellite Systems Engineering

2025 – 2026

Python, orbital dynamics

- Contributed to Team Anant's orbit-propagation and spacecraft-analysis sandbox, refactoring the propagator into a more modular simulation architecture.
- Developed solar-power analysis and Monte Carlo analysis workflows; contributed to telemetry, tracking, and command (TT&C) link/contact analysis and magnetorquer detumbling features.
- The simulator models orbital perturbations and spacecraft subsystems, with monitors for power generation, ground-station contact, link margin, and attitude behavior.
- Repository: github.com/team-anant/sat-sim.

Gradient-Based Antenna Array Optimisation

Research project

Aug – Dec 2025

PyTorch, EM solvers

- Developed gradient-based optimisation using array-factor models to improve antenna-array gain, beam-steering performance, and efficiency by combining physics-based modelling with deep-learning methods.
- Explored spiral-array wireless-power transfer, hexagonal-array directivity maximisation, and grid-array beam steering; compared gradient-based optimisation with differential evolution, genetic algorithms, and particle swarm optimisation.
- Repository: GitHub project.

Algorithmic Trading Competition Platform

Tech-fest competition backend

Mar – Apr 2026

Go, Python, Protocol Buffers

- Built a competition backend that handled more than 200 player strategies concurrently at a one-second market tick.
- Implemented the Go order book and matching engine, concurrent strategy evaluation, and a Protocol Buffers interface to Python strategies.
- Repository: GitHub project.

Micromouse

Personal robotics project

Raspberry Pi Pico W, MicroPython, BLE

- Built a maze-solving robot with motor encoder feedback, an MPU6050 IMU, and three ultrasonic distance sensors.
- Developed MicroPython firmware and a Python BLE client for monitoring sensor readings, sending commands, tuning navigation parameters, and transferring JSON logs.
- Repository: github.com/ProAltro/micromouse.

Huffzip

Personal compression project

C++

- Implemented a file compression and decompression tool combining optional LZ77 precompression with Huffman coding.
- Designed a compact format with CRC-32 integrity checks and verbose coding statistics.
- Repository: github.com/ProAltro/huffzip.

Additional DSP and Platform Projects

2023 – 2026

Selected course, club, and event work

Python, Go, Next.js

- Co-developed an audio watermarking system using chaotic encryption, three-level Haar DWT, and DTMT; refactored embedding/extraction modules and tests. On the tested track, watermark correlation was 0.993 under 20 dB Gaussian noise and 0.998 after 128 kbps MP3, with marked sensitivity to time scaling. Project.
- Built a Go/Gin/SQLC authentication service for the BITS Postman recruitment platform, including Google OAuth sessions, token verification, logout, and club-head login.
- Developed Team Anant's website and contributed to its internal Flask ERP for project, inventory, finance, and user/role management workflows.
- Built the BITS Pilani Qiskit Fall Fest 2026 website around the official programme: 8 days, 16 sessions, and 27 taught hours, with interactive and responsive presentation.

Leadership and Campus Involvement

Systems Engineer - Team Anant, BITS Pilani satellite club

May 2024 – Present

- Coordinate subsystem interfaces for a 3U CubeSat mission.
- Work across TT&C link budgets and modulation schemes, power budgeting, OBC state transitions, and ADCS state estimation with Kalman filtering; consider magnetorquer actuation and payload constraints in mission design.

Cultural Fest Coordinator - Matrix, The Film and Literature Club

May – Dec 2025

- Organised the book fair and ran film screenings and quizzes as part of the cultural fest.

IEEE BPSC (AI/ML Vertical); Coding Club (competitive programming); The Debating Society.

Technical Skills

Programming: Go, Python, C++, SystemVerilog

Systems: Linux, networking, eBPF, concurrency, Protocol Buffers

Hardware: RTL design, NoC and GPU interconnects

Machine Learning and Optimisation: PyTorch, GANs, diffusion models, gradient-based optimisation

Selected Coursework

Taken beyond my core discipline out of interest: Deep Learning; Quantum Informatics and Computing; Special Relativity; Group Theory; Topology; Cryptography.

Additional Information

Career update: Placement offer from Graviton Research Capital.